DIGITAL TWIN FOR UNDERGROUND STATIONS: IMPROVING DECISION MAKING FOR CONSTRUCTION LIFECYCLE

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ABSTRACT

Challenges in the extraction and use of earth resources and spaces are encountered given a growing worldwide population, rising infrastructures development, and widespread climate change. In Australia, mining and construction are two major bases for economic growth while both being traditional hazardous and heavy industries. A nation-wide infrastructure upgrade featuring large-scale underground development is underway, the geological uncertainties and localisation difficulties of already laid infrastructure are associated with challenges not seen in building construction. A safer and competent subterranean transport solution is yet proposed in the context of sustainable developments. In light of this, geotechnical analysis as a fundamental subject for developing and maintaining safe and sustainable use of underground space has huge potential to be undertaken more intuitively considering the advancements in information management and visualisation. The PhD work examines the state-of-the-art applications, limitations and future opportunities of Building Information Modelling (BIM) and other computational techniques in the digitisation of tunnelling and underground construction. The visualisation and interoperability facilitated by data-driven processes are especially important to underground construction that engages interdisciplinary and multi-environment interaction.

1 INTRODUCTION

After decades of development, digital engineering tools, particularly BIM and associated processes and standards have been regularly used in the architecture, engineering and building (AEC) industries (DAUB, 2019; Eastman et al., 2011). A key enabler of this transition is how information is handled, which includes both geometries and properties. Correct use of information and exchange mechanism yields efficiency that eventually converts to smooth project delivery and cost savings. In Australia, a nation-wide infrastructure upgrade featuring large-scale underground development is underway, transport sector is especially vulnerable to risks due to 25% increase of tunnelling projects in the next few years, which leads to critical issues, majorly on tunnelling and ground movement (Infrastructure Australia, 2021). The research focuses on improving efficiency and interoperability of cross-disciplinary and cross-phase applications for station infrastructure, characterised by wide-span, complex structural form, and settlement-sensitive properties. We propose a multi-module system that coordinates geographic information system (GIS), geological interpretation, excavation modelling and BIM for a variety of engineering applications in underground infrastructure projects. With the use case of geotechnical analysis, the multiple Level-Of-Details (LOD) BIM models feature visual programming-enabled object-oriented modelling (ground, excavation, station parts and support systems, e.g. rock bolt, lining, column) that are instantiated parametrically, both in geometrical alterations, and semantic properties (e.g. ground properties), according to specific station designs. An interactive platform that bridges information modelling and numerical modelling via automation to reduce manual efforts and cost-bearing errors is developed. The platform prototype is essentially underpinned by the digital twin concept that describe the situation where a physical entity is connected with a virtual counterpart via data communications. The research is focusing on underground stations as they are important infrastructures while possessing geometrical complexity and geotechnical challenges during construction.

Underground stations take a variety of forms corresponding to the characteristics of different construction methods. Commonly used methods include cut-and-cover, full-face tunnelling boring machine (TBM) excavation, and partial-face New Austrian Tunnelling Method (NATM) and its alternatives. The cut-and-cover method is often suitable for shallow buried underground stations, and most of the structures are rectangular frame structures or arch structures. Figure 1a-c show the cross-sectional view of the rectangular frame structure of several stations, for example, the Beijing Station of Beijing Metro Line 2. The predominant feature of this construction method is that the temporary retaining wall of the foundation pit and the side wall of the station structure are combined into one structure.

The underground stations constructed by the mining method can be divided into single-arch stations, double-arch stations and separate stations according to different geological conditions and usage requirements. Cross sections of a single arch could be consisted of one-, three, or five-centre circles with smooth transition, and the platforms are often consisted of single-, or double-column, with single-, or double-span. When space and load bearing criteria are set, single-column. Arch-based stations are divided into two basic forms: double-arch and triple-arch. With the emergence of New Austrian Law, double-arch column underground stations are gradually replaced by single-arch stations in rock formations. The three-arch and two-column structure above is widely used in the Beijing subway (Shi et al., 1997). Figure 1d-e illustrates the cross section of a typical single-arch and a three-arch and two-column structure. Figure 1f shows a split platform,

featuring a layout of a station that has a platform for each track and consists of two or more tunnels, with cross passage communicating in between and vertical staircases to connect the often double-deck structure.

The structural form of the station built by the shield method is closely related to the type of shield. The traditional shield station could be in the form of a single-circle shield, as shown in Figure 1g and connected by cross passages. Multi-circular face shield further enriches the forms of a shield station. In Japan, several innovative shield types have been manufactured, such as Double-O-Tube (DOT), Triple-Face (TF) and Multi-Face (MF), and are used to support various station designs (Moriya, 2017). Figure 1h illustrates the finishing form of a triple-circular shield station. In addition, underground stations have also been built by enlarging a shield tunnel or utilising part of the tunnel as the part of the station with the rest of the station constructed by mining method, as shown in Figure 1i (Cao et al., 2018).

After understanding the characteristics of underground stations, the key components of the structure could be established, which are important for developing the BIM-driven modelling process. The technical paper is organised as follows. Section 2 introduces the methodology and workflow of the research. Section 3 details the BIM-driven interoperability means developed to facilitate design-to-design efficiency. Section 4 discusses the future perspectives of the research. Section 5 concludes the paper. The innovations of research present herein are three-folded. First, we proposed an extension of station related systems in the primary open BIM data schema – Industry Foundation Classes (IFC). Then, geometry instantiation and property assignment are automated to facilitate modelling efficiency. Finally, an automatic information exchange mechanism between BIM and geotechnical analysis was developed and tested on several use cases. The research outputs have been preliminarily verified using the stations of Melbourne Metro Tunnel, and a wider industry application is expected for future improvements.

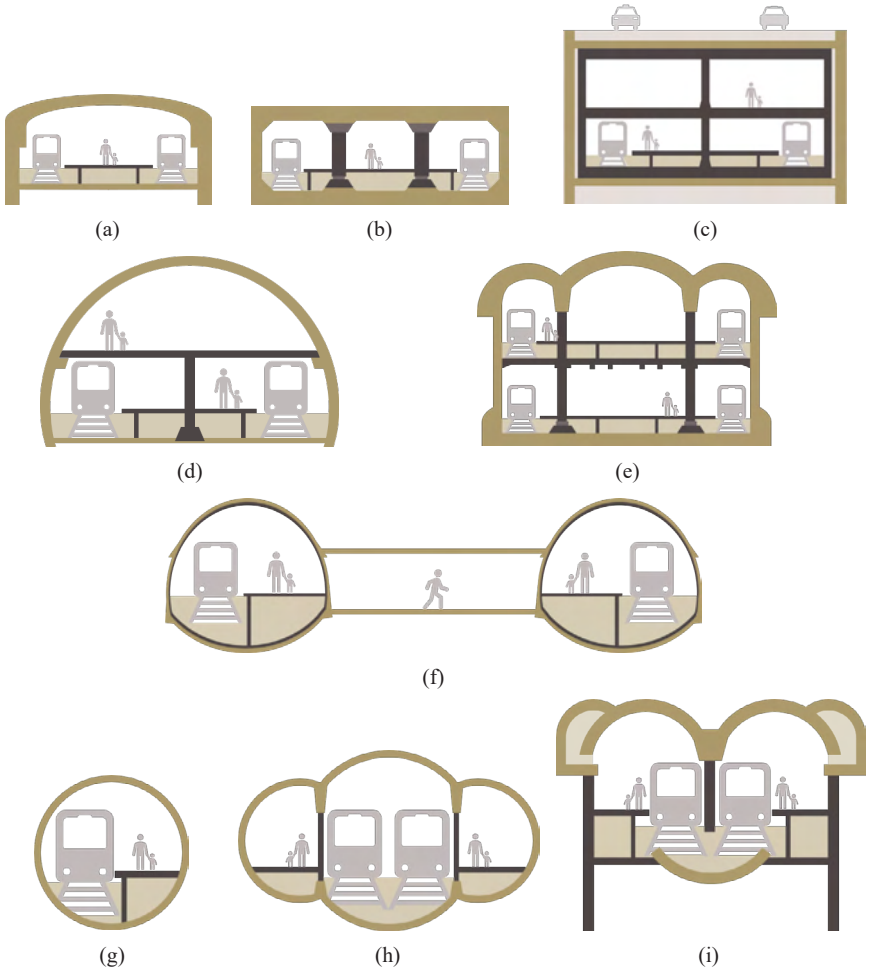


Figure 1: Schematic diagram of metro station alternative designs corresponding to different excavation methods (a-c: cross sectional view of various rectangular frame structure of different station designs; d-e: cross section of a typical single-arch and a three-arch and two-column structure; f: cross section of a split platform; g: single-shield tunnel station; h: triple-circular shield station; i: mixed shield-and-mined station)

2 METHODOLOGY AND WORKFLOW

We propose a multi-module system, as shown in Figure 2, that coordinates GIS, geological interpretation, excavation modelling and BIM for a variety of engineering applications in underground infrastructure projects. Geological modelling and interpretation for numerical analysis of geotechnical works have been a routine practice. While BIM has been increasingly employed for project design authoring, construction monitoring and operation management, the exchange channel between BIM and numerical modelling has yet been established. Various information sources and associated techniques are engaged in underground construction lifecycle. The differences of scale, information category, and processing algorithms featured by these systems need to be addressed for better coordination among cross-disciplinary applications. Moreover, since there are various ways of establishing interoperability between heterogeneous applications, exchange mechanisms should be established in a way that allows for certain level of extensibility and adaptability. The methodologies undertaken expects to perform BIM-driven life-cycle analysis, consisting of (1) object-based modelling, (2) model-based collaboration, and (3) asset-based decision making.

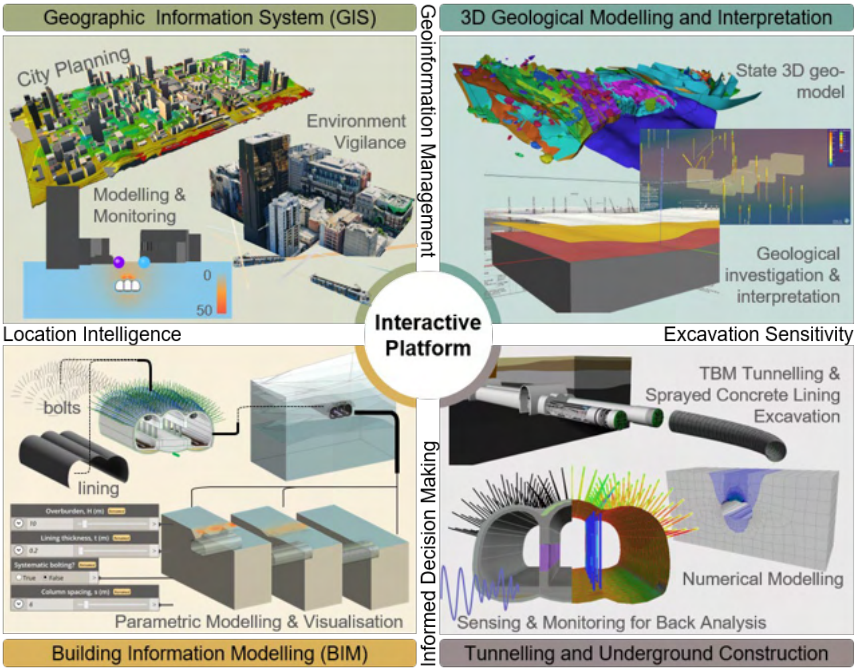


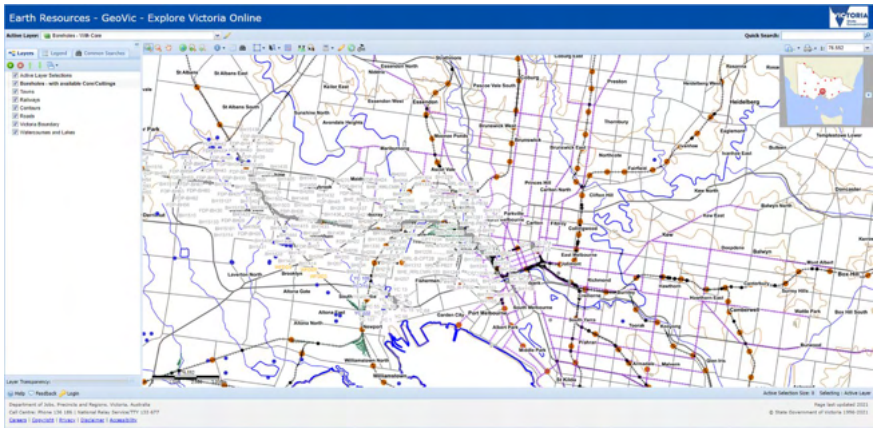
Figure 2: Interactive platform featuring four modules required by tunnelling and underground stations

2.1 GIS AND GEOLOGICAL MODELLING

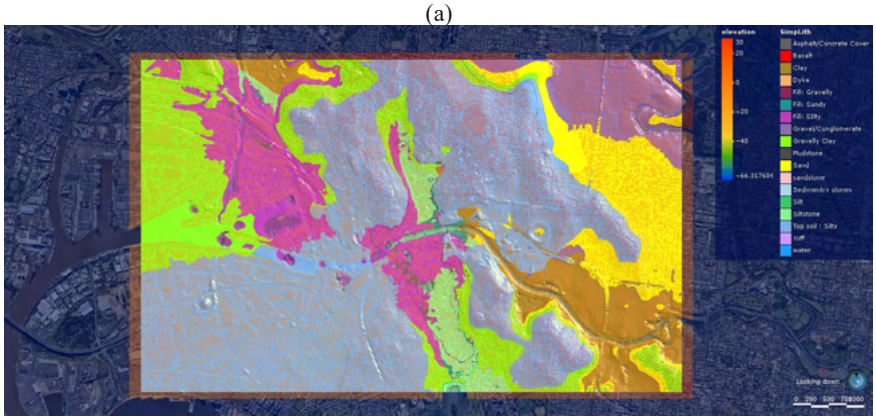
Digital transformation in urban underground development is facing unique challenges caused by the intrinsic complexity associated with spatial opaqueness, geological uncertainties, high-risk working environment and ground-machine-structure interactions. Excavation carried out without a reasonable understanding of the ground features can be especially problematic. A project-scale geological understanding established incorporating the GIS databases, geological observation and geophysical investigation is essential to mitigate the uncertainties inherited from the ground. Therefore, modelling with close reference to geographical and geological information is an important strategy for BIM in underground construction. The accuracy of the federated BIM model is to be continuously enhanced with domain-specific knowledge reconciled with as-built and as-damaged information (Huang et al. 2021).

A number of online GIS-based application platforms are maintained at different administrative levels. For example, AURIN, the Australian Urban Research Infrastructure Network, provides integrated GIS, data release and delivery service for researchers, industry and all levels of government. While interface such as AURIN provides interactive geospatial map of existing open public data, reflecting multi-categorical information (e.g, geology, geophysics, national satellite images), it does not encompass all open public data required for modelling such as borehole information, which is provided in a different interactive map under GeoVic as shown in Figure 3a. This interactive online facility displays location and depth of boreholes, and allows borehole content to be exported in several formats as separate downloads. Moreover, the water depth and ground water data of Victoria can be found in another GIS interface Visualising Victoria's Groundwater. Meanwhile, the digital elevation or terrain datasets are facilitated by the Geoscience Australia's Elevation Information System (ELVIS) to help build ground elevation models.

Once the data are collected from respective sources, there are two main approaches used for 3D geological modelling. The constructive cross-section approach, which is an explicit method driven by geological concepts and knowledge, where geological surfaces are created by interpreting geological maps and correlating cross-sections over sparse data. The geostatistical approach, on the other hand, is an implicit method that has been prevalently incorporated in commercial software, where mathematical functions based on geological fundamentals are utilised to form numerical interpolation and extrapolation of fractural geological and geophysical data. Supported by topological data obtained from ELVIS, a 3D geological model is shown in Figure 3b, where the borehole data are discretised into scatter points and interpolated to determine strata elevations and form the final model constructed with 3D blocks.



(a)



(b)

Figure 3: (a) GIS-based "borehole" database; (b) 3D Geological modelling

2.2 BIM-DRIVEN DESIGN-TO-DESIGN INTEROPERABILITY

BIM can be differentiated into "Closed BIM" and "Open BIM" that are distinguished by the implementation of open, neutral data exchange formats (Borrmann et al., 2018). The common understanding of the language used in the product model is becoming increasingly important to ensure the consistency of information exchanged and to facilitate the multidisciplinary collaboration in major infrastructure projects. In this research, interoperability between structural design and geotechnical design is explored between two interfaces Autodesk Revit (2019.2) and Itasca FLAC3D (Version 7.0).

There are several numerical modelling formulations based on continuum and discontinuum approaches. FLAC3D belongs to the former, which is underpinned by finite difference method, and thus requiring meshes to be created for the mathematical solutions to a set of partial differential equations, relating mechanical and kinematic variables, to be reached at the vertices of the mesh. Therefore, the first module of the exchange enabler is dedicated to specifying meshing requirements and grouping areas of interest. Instead of using the built-in FLAC3D methods (primitives, extrusions, and building blocks) to generate zones for grids, the module employs an additional zone generation functionality, referred to a geometry-based densification in FLAC3D that allows direct import of the geometry established in Revit STL (i.e. stereolithography) format, or the DXF (Drawing Interchange Format) which are file formats supported by many software systems for representing triangulated surfaces. Hence, the relevant BIM-based geometric model can be directly imported, instead of re-modelling the entire geometry in the FLAC3D interface.

The next module facilitates specification of initial conditions, boundary conditions, and assigning constitutive behaviours and properties leveraging information stored in both the BIM models and project database. A third module allows for

setup of a process-oriented simulation that includes excavation of materials and installation of ground support. Due to the flexibility of the exchange enabler, different simulation scenarios can be easily implemented by modifying the inputs or selecting certain options in the user interface. Finally, a module is set up for automating the program execution. This suggests that users are able to call the program without entering the interface.

Dynamo, underpinned by visual programming, runs pre-composed scripts to generate models. We extended this automation process by embedding python scripts in dynamo scripts, and thus on top of automation, data are stored, exchanged and manipulated into an interpretable form by leveraging the object-oriented scripting and programming functions of the Python language. These files produced and/or handled by dynamo and python are mainly in two formats, the stereolithography (.stl) for representing triangulated surfaces (geometry data), and executable files, which are called in FLAC3D carrying out numerical simulation (containing semantic data). Different dynamo and python scripts are combined into "modules" depending on the four purposes: (1) produce .stl-based geometry and create executable file performing meshing and grouping; (2) create executable file specifying initial conditions, boundary conditions and assigning constitutive behaviours and properties; (3) create executable file controlling excavation and support installation, and (4) create a batch file that can be interpreted by a command-line interpreter to automatically run the FLAC3D program and commence simulation. The exchange enabler is made to link design modelling and the numerical simulation via automation that is achieved by a visual programming tool and a programming language. During the exchange process, manual handling of data is not required as all scripts are automatically run and interpreted to produce files required for the next step in the workflow, thus reducing chances of errors. The overall data exchange flow is illustrated in Figure 4.

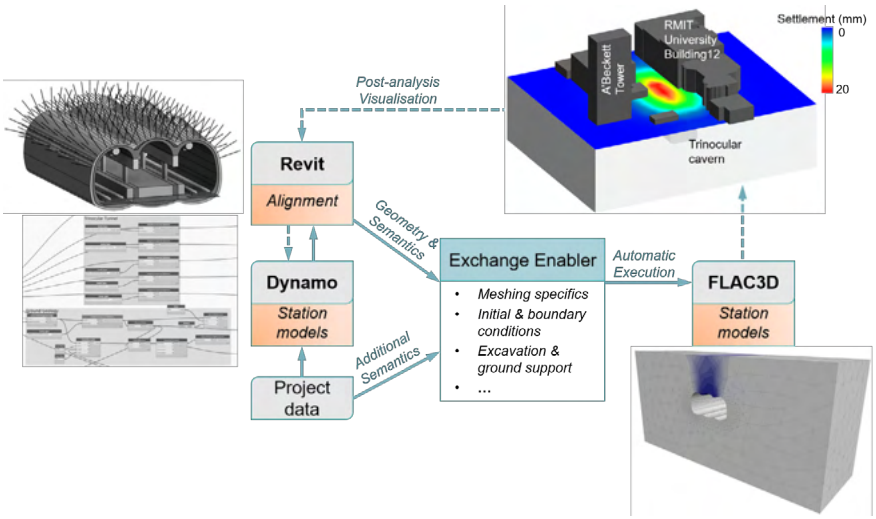


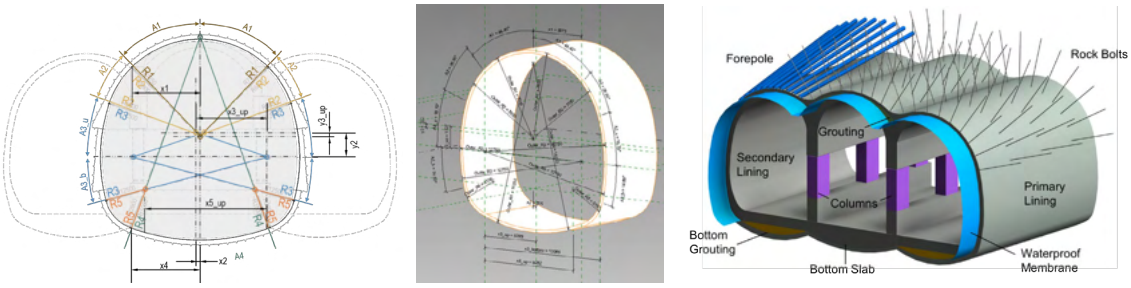
Figure 4: Flowchart of information exchange between BIM and numerical modelling (modified after Huang et al, 2022)

2.2.1 Parametric model

The object-oriented parametric modelling is at the core of the BIM technology, with 'parametric' describing an automated process of adjusting adjacent element or assembly to maintain a predefined relationship with an element modified (Azhar et al., 2012). Different from mechanised tunnelling that creates cylindrical space sealed with standard lining segments along its circumference, with its BIM design and analysis potentials examined by (Ninić et al., 2020a, Ninić et al., 2019, Ninić et al., 2020b), in conventional tunnel method the profiles of the main structures (e.g. large-span trinocular cavern) generated by sequential excavation are often non-standard and specific to individual projects.

The preliminary design of the State Library Station is used to demonstrate the parametric assembly modelling of the station complex. The station platform consists of a trinocular cavern. Based on the dimensions given to the cross section of the cavern, we first established a parametric 2D profile of the centre tunnel following the construction line (Figure 5a) representing the structure after primary lining is installed. The five centre circle outline is controlled by a set of constraining operations engaging trigonometric identities and equations. The variables include the respective radius and angle of the circles constructing the tunnel. These dimensional parameters are considered associated with the component type, thus are accessible for modification in "Family Types". By locking the dimensions, constraints are created to restrain the movements of the referenced curve elements.

After establishing the shapes, the solid component is built by solidifying based on the model line representing tunnelling, with the "Solid Form" command and extrude orthogonally. Later, "Void form" command is used based on the model line representing lining being extruded along the same direction (Figure 5b). Figure 5c shows the complete model assembly.



(a) (b) (c)

Figure 5: (a) Parametrically controlled profile generation; (b) solid modelling engaging extrusion; (c) parametric assembly of solids

2.2.2 IFC data model extension

Interoperability can be facilitated by developing standards or by developing bridges between two applications (Wegner, 1996), with the latter being dominant in the existing researches while the former approach for applications used in geotechnical analysis is limited. The standardisation of information model via IFC could be promising as interoperability relies on specified data formats (Ide and Pustejovsky, 2010).

An official IFC data model for underground infrastructure is yet to be developed. To facilitate the representation of the exchange requirements (both geometry and semantic requirements), an extension to IFC schema is proposed to assist the implementation of MVDs. A taxonomy analysis for underground stations is applied to help scope out the extension, which is shown in Table 1. It is worth noting, station is herein classified as a built facility along with other physical infrastructures such as bridge, port, rail and tunnel, in accordance with the taxonomy of buildingSMART IFC extensions for infrastructure.

The extension is separate into spatial elements and physical elements. The spatial elements captured the spatial hierarchy of a project (Borrmann et al., 2019). Three subclasses of spatial elements are proposed for the station, including IfcStation, IfcStationPart, and IfcStationExcavationZone. The first two were inherited from IfcFacility and IfcFacilityPart that were added to the released candidate IFC schema to accommodate possible future definitions of built facilities (buildingSMART, 2020). The types of station named after construction methods include cut and cover, drill and blast, TBM or shield tunnelling, and sequentially excavated. Majority of physical element additions can be defined through the existing entities. The physical entities newly defined specifically for the station are IfcLining and IfcForePoling as subtype of IfcBuildingElement, and IfcRockBolt as subtype of IfcReinforcingElement. Figure 6 illustrates the physical extensions by means of a Unified Markup Language (UML) model with added elements marked in red.

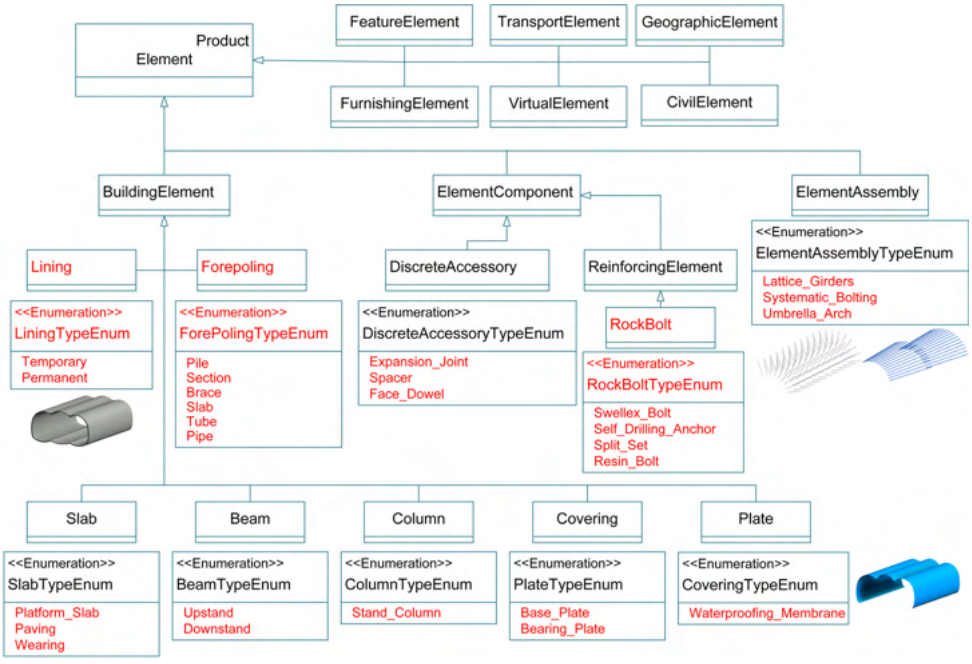


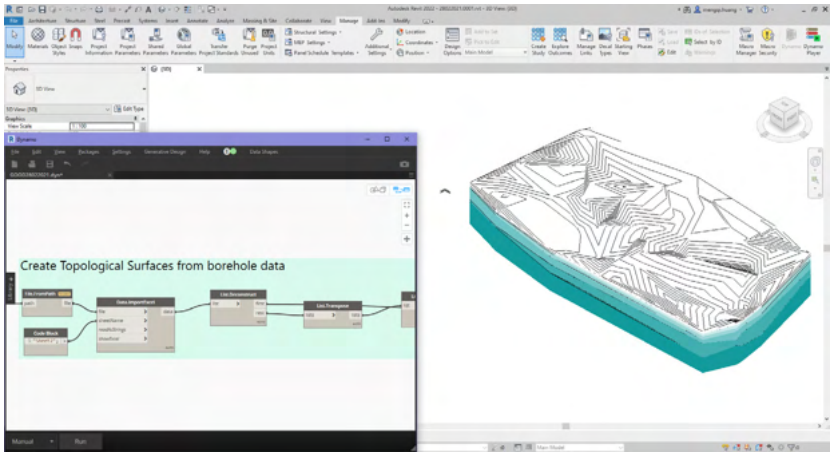
Figure 6: IFC Extension (modified after Huang et al., 2022)

2.3 REVIT GEOLOGICAL MODEL

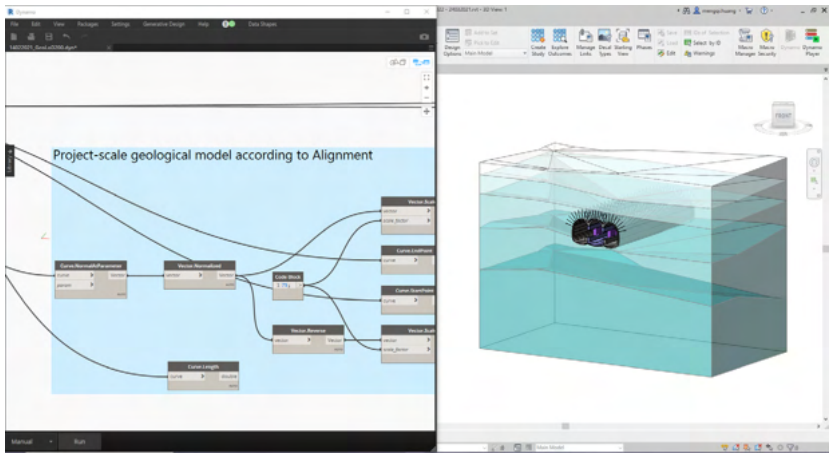
An algorithm is developed to automatically retrieve and combine the geological data given in form of the borehole data to model and characterise the subsurface conditions. The process retrieves borehole data (synthetic) stored in an Excel file format and uses topological relations to construct a simple stratified geological model (Figure 7a). Then the model is scaled down to the domain of numerical model, by defining a bounding box that creates a section of suitable size.

The regional model is achieved by creating a cuboid that has its y-direction parallel to the cavern alignment and truncating at a length equivalent to the length of the alignment, and then extending bidirectionally in x-direction to a width suitable for numerical modelling that ensures no boundary effects. The project-scale geological model is obtained by splitting the stratified geological model along the circumferential surface of the cuboid (Figure 7b). This process preserves the strata corresponding properties for later. It is worth mentioning that this stratified geological model is created to demonstrate the feasibility of the workflow, which focuses on BIM-numerical modelling interoperability, the limitations of this modelling approach in regard to treating uncertainties of geomodelling or geological interpretation provided by specific discipline modelling formulations are acknowledged.

Assembly of the information models, i.e. parametric geometries and semantic embedment using computational design approach is achieved by three main steps. First, the excavation (represented by the void) is modelled, which involves determining whether the modelled geo-bodies have intersected with the TobeExcavated element and splitting the intersected geometries along the surfaces of the TobeExcavated element. Second, parametric objects based on extrusion, such as outer shell and waterproofing membrane, are instantiated along the alignment. Finally, parametric objects based on face division such as columns, bolts and canopy tubes are instantiated on selected faces. All the aforementioned procedures are automated using dedicated algorithms. Two dedicated algorithms are designed to add shared parameters to model components, retrieve material strength data stored in another Excel Spreadsheet and assign the value to corresponding parameter category.



(a)



(b)

Figure 7: Parametric modelling of (a) geological body and (b) intersecting and perforated by a proposed construction

3 BIM-DRIVEN DESIGN-TO-DESIGN INTEROPERABILITY

The interface selected for numerical simulation for geotechnical analysis is FLAC3D, which supports a command-driven program execution and allows the use of data files (.dat format) to describe the model. This supposedly outdated program design is actually favoured in information exchange scenarios as it makes the complete automation of program execution possible.

An exchange enabler is proposed to facilitate exchange of BIM model and automate the execution of numerical simulation. The enabler is realised by a set of Python scripts under a modular structure to allow future extensions and convenient modifications. The modules are constructed in accordance with the general solution procedure of the numerical modelling program. The framework of the proposed workflow and processes linked by the exchange enabler are illustrated in Figure 9.

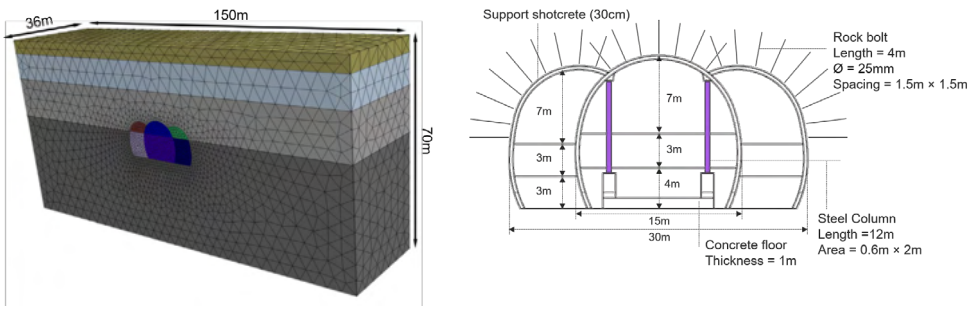
There are two approaches for importing the design geometries (when they are composed of polygons forming a closed volume) to the numerical modelling environment depending on which format the geometries are saved. The first approach imports the geometry in DXF and extrude the geometry before generating the zone from the extruder. The second approach imports the TobeExcavated and the ground model of STL files as individual volumes to the interface and perform meshing accordingly. Both approaches generate meshed models as shown in Figure 8a. Example uses unstratified ground model.

The initial condition is set in accordance with where excavation is carried out in the real world. The station is relatively shallow buried with an overburden of 25m, the vertical in-situ stress is assumed to only come from the overburden; the horizontal stresses are calculated based on the in-situ lateral stress coefficient (k_0) in the range of 0.5-2. The ground is mainly consisted of interbedded siltstone and sandstone deposited during the Silurian Period, and termed Melbourne Formation (MF) (Neilson, 2018). FLAC3D undertakes a dynamic calculation loop supported by an explicit finite volume formulation that can capture the complex behaviours of a model. The FLAC3D built-in Mohr-Coulomb failure criteria with tension cut off is utilised for describing the behaviour of rock mass, considered as an isotropic homogeneous material, under compression condition. The physical and mechanical properties of the rock are reported in Table 1 corresponding to weathering condition from extremely weathered (MF4) to fresh/slightly weathered (MF1).

The construction takes the shape of a triple arch cavern and excavation follows the NATM method. The excavation cycle begins with road header advancing 1.5m, followed by installation of support system, consisted of rock bolts, shotcrete and lattice girder (which is not explicitly present in the numerical model but its equivalent strength is considered when combined with shotcrete and simulated by the shell structural element). Sequential excavation is taken place, with the excavation of the central cavern divided into top heading, bench and invert; side caverns divided into top heading, bench and invert, as shown in Figure 8b.

Table 1: Physical and mechanical properties for MF1 to MF4

	Unit weight γ (kN/m^3)	Rock mass modulus E_{rm} (MPa)	Poisson's ratio v	Friction angle ϕ (°)	Cohesion c (kPa)	Depth (m)
MF4	22.5	100	0.3	30	37	0-5
MF3	23	200	0.25	34	60	5-15
MF2	24	800	0.2	43	115	15-30
MF1	25.5	2500	0.2	54	270	30-70



(a) (b)

Figure 8: (a) 3D numerical modelling; (b) schematic of support system and corresponding geometrical properties

A range of numerical investigations and parametric studies can be facilitated by adopting numerical simulation. However, the visualisation of the simulation results without a proper context may not be sufficient to assess the potential environmental, social and economic impacts of the project.

A reverse loop, also realised by a dedicated Dynamo algorithm, is therefore designed to help visualise the impacts of implementing current design on the surroundings. The intention behind this process is to maximise the benefits of geotechnical numerical simulation via effective visualisation. For example, the technical feasibility of underground construction is often assessed by taking the potential effects of excavation induced settlement as a criterion. Therefore, an algorithm is developed to reads the simulation results of displacement at nodes on the ground surface and displays them in BIM software viewer using colour-filled contour representation, which is mainly achieved via a built-in node namely "FaceAnalysisDisplay" and augmented surface deformation that is achieved via "Topology.ByCoordinates". Figure 9 shows the effect visualisation using the algorithm based on a parametric study. Three parameters are investigated for their impact on ground surface displacements including overburden depth, lining thickness and presence of column. In addition, during the construction phase, the same BIM model with the proposed viewer can be used to store and visualise monitoring data such as ground settlements. In that case, the proposed visualisation algorithm also allows for comparison of predicted settlement against monitored settlement and highlights locations of where the difference exceeds acceptable limits that might lead to increased risks during the construction.

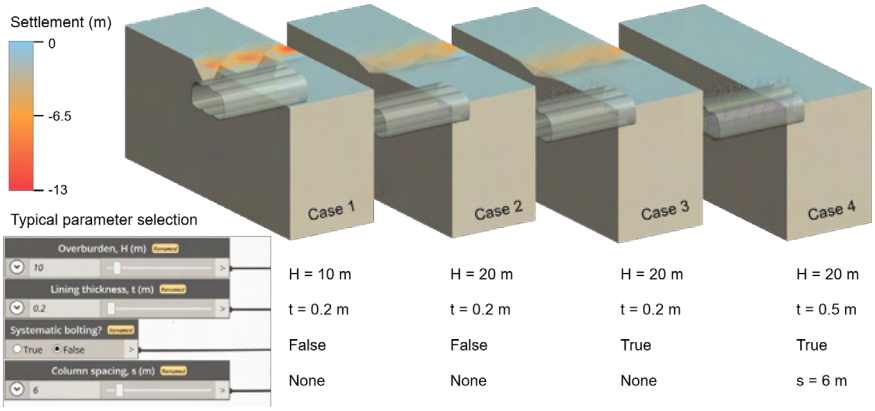
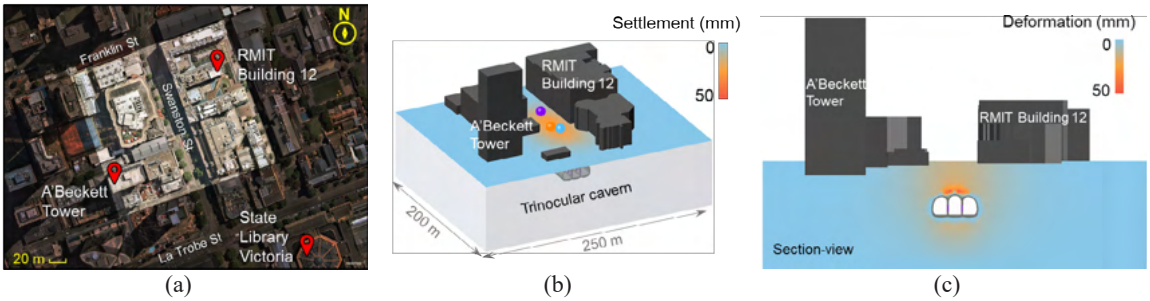


Figure 9: Numerical analysis visualisation parametric study with variations in overburden depth, lining thickness and presence of columns (Case 1 with Overburden H=10 m, Lining thickness t=0.2 m, Systematic bolting: False, Column: False; Case 2 with H=20 m, t=0.2 m, False, False; Case 3 with H=20 m, t=0.2 m, True, False; Case 4 with H=20 m, t=0.5 m, True, Column spacing: s=6m)

Figure 10 illustrates the proposed visualisation algorithm on the design prototype based on the platform cavern of the Melbourne Metro State Library Station and the city context of the relevant region. This visualisation node forming part of the workflow incorporates both BIM and GIS to demonstrate the benefits of effect quantification and visualisation to facilitate informed decision making, this is especially useful when the environment is sensitive to excavation induced settlement. For example, settlement accumulation throughout the construction are monitored. Three of the many surface monitoring points implemented along the Swanston Street are illustrated in Figure 10b. More visualisation features can be developed in future versions such as a module that can export analysis results of all components individually, including support systems, and visualises the results in a BIM model in an effective and intuitive manner, such as highlighting only parts where deformation or strain has exceeded certain acceptable limits.

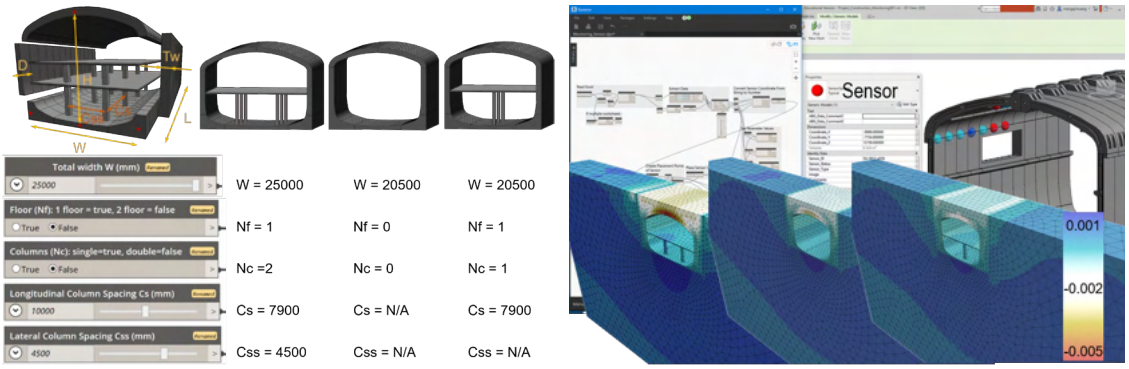


(a) (b) (c)

Figure 10: (a) geographical region of interest (photomap generated using Nearmap)**; (b) modelled settlement visualisation with three surface settlement monitoring points in their respective locations; (c) section-view of deformation visualisation**

4 FUTURE PERSPECTIVES

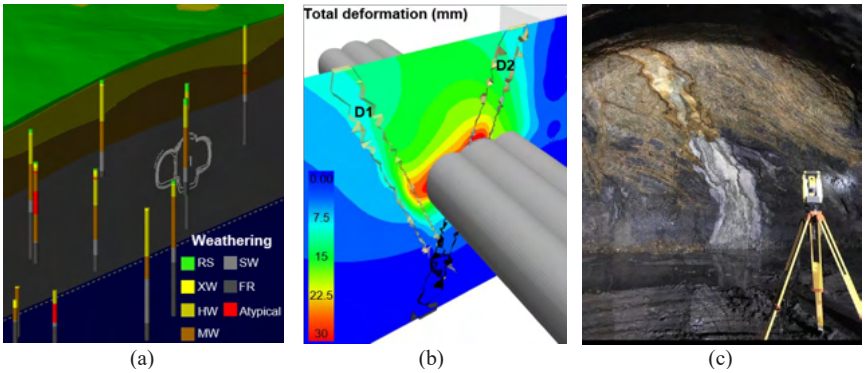
Sustainable development of underground construction requires adoption of digital engineering and emerging technologies. Research interest and engineering practice of modular construction for underground station have increased over the past few years (Yang and Lin, 2021). The parametric design for prefabricated station was explored, especially for project planning and preliminary design stages leveraging the methods developed in this study. Figure 11a demonstrates the parameterisation of segmental composition of a prefabricated station taking the Parkville cut-and-cover metro station geological and spatial requirements as an example. Figure 11b illustrates the use case of numerical analysis. Based on the numerical simulation and in-situ monitoring, design optimization through back-analysis can be carried out in BIM environment forming an engineering decision-making loop to further improve efficiency.



(a) (b)

Figure 11: (a) Parametric modelling driven generative design for prefabricated metro station; (b) 3D numerical model and simulation result, and digitised in-situ monitoring result

Further investigations were made to integrate 3D geological modelling and GIS databases with BIM to allow different levels of control over design, improve digitisation and specify visualisation requirements of the ground. As shown in Figure 12, model of the ground, boreholes and station structure are present in their respective real-world locations, where their geographic relationships are built to provide context for further analyses. Geological model with high level of confidence help engineers identify risk zones and aid in decision makings of excavation and support measures; in particular, the identification of geological discontinuity near the structure would be imperative to the stability of excavation. Figure 12b shows the numerical modelling of two dykes (D1 & D2) one coming near the trinocular arch and one intersecting with the arch in order to understand their effects on the construction. Corresponding geological features were discovered on site as shown in Figure 12c (Rail Projects Victoria, 2022), suggesting the importance of studies into interoperability, data transparency and investigations among the stages of geo-investigation, design and construction.



(a) (b) (c)

Figure 12: (a) Combined model of ground, borehole and structure; (b) numerical modelling of geological features intersecting structure; (c) site observation of geological feature (Rail Projects Victoria, 2022)

5 CONCLUSIONS

With a digital transformation agenda expanding into the domain of underground, we examined the state-of-the-art applications and future opportunities of BIM that are believed to demonstrate huge potential in the digitisation of tunnelling and underground construction. BIM enables the information collection, exchange and linking throughout a project's lifecycle. The visualisation and interoperability facilitated by BIM processes are especially important to underground construction that engages interdisciplinary participation and multi-environment interaction. The geological

uncertainties and localisation difficulties of already laid infrastructure are challenges not seen in building construction. Underground BIM thus requires consideration of geographical and geological features, and harnessing data exchange solutions to establish a comprehensive and integrated information resource.

This study proposed an interactive platform that considered the comprehensive and collaborative integration of GIS, 3D geological modelling, construction methods, and sensing technologies into BIM in order to form a reliable basis for decisions and management during the lifecycle of an underground project. The presented work shows that there is a strong potential of using advanced information modelling and automatic workflow to streamline the processes and mitigate workload of the engineers while protecting data accuracy and usability.

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